

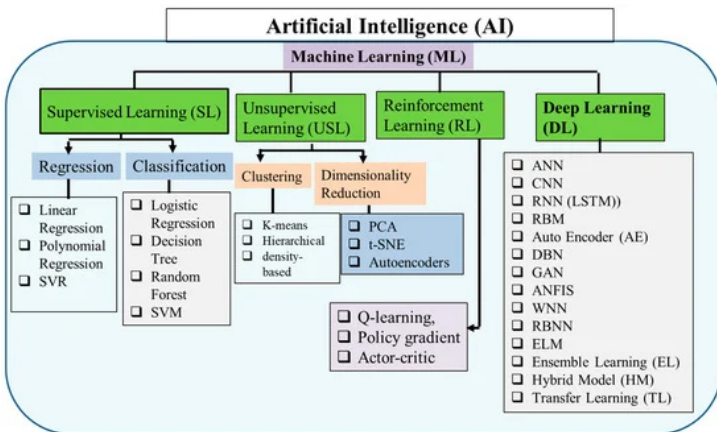
# Comparing Logistic Regression and Bayesian Network for Predictive Modeling

Bhakti Patel  
Advisor: Dr. Ivan Dungan

Francis Marion University  
Undergraduate Mathematics Conference 2024

April 12, 2024

# What is machine learning?



<https://www.mdpi.com/2071-1050/15/9/7087>

# Project Goal

- Predict customer's status for future year.
- Handle imbalanced data using:
  - Under Sampling.
  - Over Sampling.
- Compare the two models (Logistic Regression and Bayesian Network).
  - Train and Test on the 2022 Data.
  - Compare with 2023 Data and perform evaluation metrics.

# Dataset and the variables

- **Dependent/Target Variable :**
  - Status (Active or Not Active)
- **Independent/Predictor Variables :**
  - Product 1
  - Product 2
  - Product 3
  - Product Type
  - Years Owned
  - Size of Clients

# Project Background - Bayesian Network

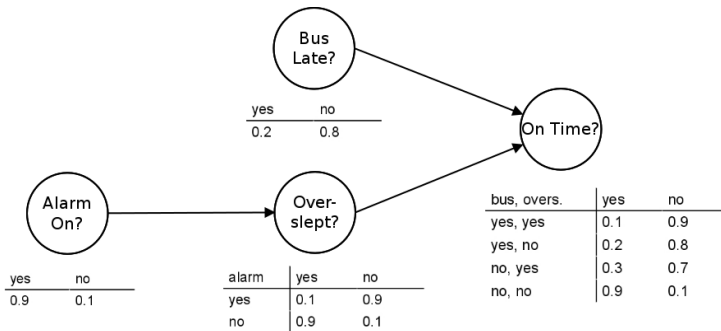
- **Probabilistic Graphical Model** comprised of nodes and directed edges.
- Provides a way of applying **Bayes Theorem** to complex problems.

$$P(A|B) = \frac{P(A) \times P(B|A)}{P(B)}$$

where,

- P(A) is Probability of event A.
- P(B) is Probability of event B.
- P(A|B) is Probability of event A given event B.
- P(B|A) is Probability of event B given event A.

# Bayesian Network Example



<https://www.uib.no/en/rg/ml/119695/bayesian-networks>

# Project Background - Logistic Regression

- Statistical method used for **binary classification tasks**, predicting the **probability** of a binary outcome.
- Extends linear regression by using the **logistic function** to model the relationship between **independent variables** and **log-odds** of the outcome.

$$P(Y = 1|X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n)}}$$

where,

- $P(Y = 1|X)$ : Probability of the status being "Active" given the predictors.
- $\beta_0, \beta_1, \dots, \beta_n$  are the coefficients estimated by the model.
- $X_1, X_2, \dots, X_n$  are the predictor variables.

# Logistic Regression Example



<https://www.natasshaselvaraj.com/logistic-regression-explained-in-7-minutes/>



# Data Cleaning and Preprocessing

- One - Hot Encoding for Logistic Regression.

| Original Data |        | One-Hot Encoded Data |        |        |        |
|---------------|--------|----------------------|--------|--------|--------|
| Team          | Points | Team_A               | Team_B | Team_C | Points |
| A             | 25     | 1                    | 0      | 0      | 25     |
| A             | 12     | 1                    | 0      | 0      | 12     |
| B             | 15     | 0                    | 1      | 0      | 15     |
| B             | 14     | 0                    | 1      | 0      | 14     |
| B             | 19     | 0                    | 1      | 0      | 19     |
| B             | 23     | 0                    | 1      | 0      | 23     |
| C             | 25     | 0                    | 0      | 1      | 25     |
| C             | 29     | 0                    | 0      | 1      | 29     |

<https://www.statology.org/one-hot-encoding-in-python/>

- Under Sampling:
  - Reduce the size of the majority class to balance the class distribution.
- Over Sampling:
  - Increase the size of the minority class to balance the class distribution.

# Confusion Matrix

|           |            | Actual |            |
|-----------|------------|--------|------------|
|           |            | Active | Not Active |
| Predicted | Active     | TP     | FP         |
|           | Not Active | FN     | TN         |

where,

TP = True Positive

FP = False Positive

FN = False Negative

TN = True Negative

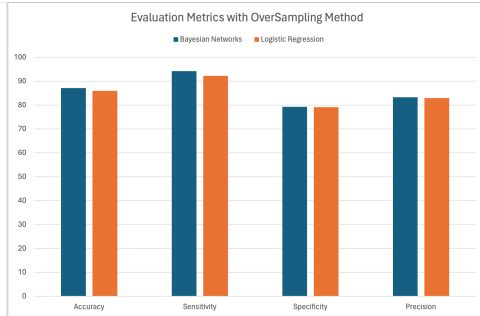
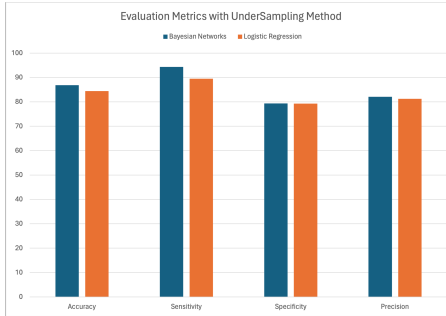
# Evaluation Metrics

- **Accuracy** (Correct Prediction Rate) =  $\frac{TP + TN}{TP + TN + FP + FN}$
- **Sensitivity** (True Positive Rate) =  $\frac{TP}{TP + FN}$
- **Specificity** (True Negative Rate) =  $\frac{TN}{TN + FP}$
- **Precision** (Positive Predictive value) =  $\frac{TP}{TP + FP}$

# Results without sampling methods



# Results with sampling methods



# Communicating with the models

- In Bayesian network one can **query** the model.

$P(\text{Status} \mid \text{Product 2} = \text{False})$

| Status             | phi(Status) |
|--------------------|-------------|
| Status(Active)     | 0.6727      |
| Status(Not Active) | 0.3273      |

$P(\text{Status} \mid \text{Product 2} = \text{True})$

| Status             | phi(Status) |
|--------------------|-------------|
| Status(Active)     | 0.9479      |
| Status(Not Active) | 0.0521      |

- Logistic Regression identifies the **strength** and **direction** of the relationship between predictors and the outcome.

| Logit Regression Results |                  |                   |          |       |           |          |
|--------------------------|------------------|-------------------|----------|-------|-----------|----------|
| Dep. Variable:           | Status           | No. Observations: | 8088     |       |           |          |
| Model:                   | Logit            | Df Residuals:     | 8078     |       |           |          |
| Method:                  | MLE              | Df Model:         | 9        |       |           |          |
| Date:                    | Mon, 08 Apr 2024 | Pseudo R-squ.:    | 0.5215   |       |           |          |
| Time:                    | 16:19:38         | Log-Likelihood:   | -1506.9  |       |           |          |
| Converged:               | True             | LL-Null:          | -3148.9  |       |           |          |
| Covariance Type:         | nonrobust        | LLR p-value:      | 0.000    |       |           |          |
|                          | coef             | std err           | z        | P> z  | [0.025    | 0.975]   |
| const                    | 5.8194           | 2.46e+06          | 2.37e-06 | 1.000 | -4.82e+06 | 4.82e+06 |
| Product1                 | 1.4655           | 0.261             | 5.610    | 0.000 | 0.953     | 1.978    |
| Product2                 | 0.7543           | 0.151             | 4.995    | 0.000 | 0.458     | 1.050    |
| Product3                 | 0.7122           | 0.249             | 2.859    | 0.004 | 0.224     | 1.200    |
| ClientSize               | -0.0016          | 5.39e-05          | -30.235  | 0.000 | -0.002    | -0.002   |
| YearsOwned               | -1.2043          | 0.042             | -28.499  | 0.000 | -1.287    | -1.121   |
| ProdT1                   | 1.3211           | 2.46e+06          | 5.37e-07 | 1.000 | -4.82e+06 | 4.82e+06 |
| ProdT2                   | 1.6061           | 2.46e+06          | 6.53e-07 | 1.000 | -4.82e+06 | 4.82e+06 |
| ProdT3                   | 2.0558           | 2.46e+06          | 8.36e-07 | 1.000 | -4.82e+06 | 4.82e+06 |
| ProdT4                   | 0.8364           | 2.46e+06          | 3.4e-07  | 1.000 | -4.82e+06 | 4.82e+06 |

# Conclusion

- **Flexibility:**

- **Bayesian Networks:** Excel in modeling complex dependencies among variables.
- **Logistic Regression:** Offers simplicity and ease of interpretation.

- **Interpretability:**

- **Bayesian Networks:** Offer high interpretability, enabling stakeholders to understand the model's decision-making process easily.
- **Logistic Regression:** Identifies the strength and direction of relationships between predictors and outcomes.

- **Handling of Missing Data:**

- **Bayesian Networks:** Can handle missing data more effectively, enhancing robustness in real-world applications.
- **Logistic Regression:** Provides a practical approach to missing data but may require additional preprocessing steps.

## Reference(s)

- James, G. M., Witten, D., Hastie, T., Tibshirani, R., Taylor, J. (2023). An introduction to statistical learning with applications in Python. Springer.

## Libraries used in python

- Numpy
- Pandas
- Pgmpy
- SciKit Learn
- statsModel.api



# Special Thanks

- **Dr. Ivan Dungan** (Project Advisor)
- **Dr. Dan Scofield** (Feedback and Guidance)
- **ACS Technologies** (Provided Dataset and Guidance)

Thank you!

Any Questions?

## Acknowledgement

This work was supported in part by the National Science Foundation EPSCoR Program under NSF Award OIA-2242812. Any Opinions, findings and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect those of the National Science Foundation.